

Name:

Math 418 (Tully-Doyle)

Exam 1

September 21, 2026

No notes. No calculators. *Show all your work and clearly indicate your answer for full credit. Partial solutions may receive partial credit.*

1. (3 points) For each equation, classify it as linear and homogeneous, linear and inhomogeneous, or non-linear.

(a) $u_{xxt} + 2u_t + u^2 = 0$

(b) $u_{xx} - t^2 u_{tt} = t^2$

(c) $u_t + 4u_{xt} + u = 0.$

2. (3 points) Find the solution $u = u(t, x)$ to the PDE

$$u_t - 2u_x = 0$$

$$u(0, x) = \sin x.$$

Note: you need not show work here.

3. (10 points) (a) Find the characteristic variable for

$$u_t + 2u_x + u = 1.$$

- (b) Find the general solution to the PDE.

- (c) Describe the behavior of the general solution as $t \rightarrow \infty$.

4. (12 points) Consider the initial value problem

$$u_{tt} - u_{xx} + 4u_t + 4u = 0$$

$$u(0, x) = f(x)$$

$$u_t(0, x) = g(x).$$

- (a) Show that the change of variables $u(t, x) = e^{-2t}v(t, x)$ transforms this problem to the form $v_{tt} - v_{xx} = 0$, with initial condition $v(0, x) = f(x)$, and $v_t(0, x) = 2f(x) + g(x)$.

- (b) (This continues the question from the previous page). Solve the initial value problem. Hint: use d'Alembert's solution on the reduced problem.

5. (10 points) Solve the PDE

$$\begin{aligned}u_{tt} - 5u_{xt} + 6u_{xx} &= 0, \\u(x, 0) &= \cos x, \\u_t(x, 0) &= 2 \sin x.\end{aligned}$$

6. (7 points) Find and classify all eigensolutions (that is, of the form $u = e^{\lambda t}v(x)$) to the PDE

$$u_t = u_x + u.$$

(Hint: Consider the cases $\lambda = 1$ and $\lambda \neq 1$.)